

# Lab Assignment I

16/01/2026

Mark: 24

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- Write the query, execute and paste screenshots, and make a pdf. Paste the text of the query too. You need to show a screenshot to demonstrate the correctness of your query. If you have made some changes that are not visible in the direct output, display the whole table again.
  - Submit one file .pdf file containing all answers and the file name should be <roll no>\_<name>\_labassignment1.pdf
  - Write the questions before the answers.
  - Write a small justification for your solution.
  - Complete the exercise before 4.45 PM and submit the pdf in Moodle.
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## Given Schema

Book (bookID, title, genre, publishedYear)

Author (authorID, bookID, name, age)

- bookID and authorID are unique.
  - One book can have many authors.
  - Each author must be associated with a valid book.
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## Questions

### 1. Database and Table Creation (3 marks)

1. Create a database named **LibraryDB**.
2. Create the tables **Book** and **Author** with:

- Appropriate **data types**
  - **Primary keys**
  - **Foreign key constraint** (`Author.bookID → Book.bookID`)
  - **Domain constraints** (`age > 0, publishedYear > 1947, genre NOT NULL`)
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## 2. Insert Single Records (1 mark)

Insert the following records:

### Book

`(1, '1984', 'Dystopian', 1949)`

### Author

`(1, 1, 'George Orwell', 46)`

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## 3. Bulk Insert Records (2 marks)

Insert the following records using **bulk insert**.

### Book Table

`(2, 'The Hobbit', 'Fantasy', 1937)`

`(3, 'To Kill a Mockingbird', 'Fiction', 1960)`

`(4, 'The Alchemist', 'Philosophy', 1988)`

`(5, 'Harry Potter and the Sorcerer's Stone', 'Fantasy', 1997)`

### Author Table

`(2, 2, 'J.R.R. Tolkien', 81)`

`(3, 3, 'Harper Lee', 89)`

`(4, 4, 'Paulo Coelho', 76)`

`(5, 5, 'J.K. Rowling', 58)`

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#### 4. Alter Table and Update Data (2 marks)

Modify the **Book** table to add a column:

```
rating INTEGER DEFAULT 3
```

Update the ratings of the books in order:

```
(5, 4, 5, 4, 5)
```

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#### 5. Insertion in Table(6 marks)

What will happen if you execute the following command? Explain why.

```
INSERT INTO Author VALUES (1, 1, 'John Smith', 40);
```

```
INSERT INTO Author VALUES (1, 6, 'John Smith', 40);
```

```
INSERT INTO Book VALUES (135, 'The Metamorphosis', Fantasy, 1915);
```

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#### 6. Bulk Insert Using CSV (1 mark)

- Create a CSV file named **author\_insert.csv**
  - Bulk insert all records from the CSV file into the **Author** table  
(Hint: Use LOAD DATA INFILE / COPY / DBMS-specific command)
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#### 7. Queries (1 mark each)

1. Display all records from the **Book** table.
2. Display only the **names and ages** from the **Author** table.

3. Display names of authors who are **older than 60**.
  4. List titles of books **published after 1990**.
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#### 8. Update based on Condition (2 marks)

- Increase the age of all authors who wrote **Fantasy** books by **1 year**.
  - Display the **Author** table after update.
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#### 9. Backup and Deletion (1 mark each)

1. Dump the **LibraryDB** database into a `.sql` file and include that in your submission.
  2. Delete the **Author** table and list all remaining relations in the DB.
  3. Delete the **LibraryDB** database and list all existing databases.
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## Practice Extension Questions

#### 10. Restore the Database

- Restore the database (from the `.sql` dump file).
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#### 11. New Table Creation

Create a table **Publisher** with:

`Publisher (publisherID, name, country)`

- `publisherID` → Primary Key

Insert:

```
(1, 'Penguin Books', 'UK')  
(2, 'Bloomsbury', 'UK')
```

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## 12. Modify Book Table

- Add `publisherID` to **Book**
  - Create a **foreign key** referencing **Publisher**
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## 13. Review Table and CSV Bulk Insert

Create a table **BookReview** with:

```
BookReview (  
    reviewID INTEGER PRIMARY KEY,  
    bookID INTEGER,  
    review VARCHAR(255),  
    rating INTEGER,  
    reviewDate DATE,  
    FOREIGN KEY (bookID) REFERENCES Book(bookID)  
)
```

- Create a CSV file with at least **5 review records**
  - Perform a **bulk insert** into the `BookReview` table
- .....